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Electronic Practical Work

Microcontrollers Security

Write Memory Protection

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Without protection, it is possible to write to the Flash (outside the Flash where the program is running). In order to improve the robustness of a program, it may be necessary to block access to unused memory sections.

SRAM WRITE PROTECTION

There are two sections of SRAM available on the STM32L476 SRAM1 and SRAM2. Only SRAM2 has write protection.

Work to be provided

- Find the address of SRAM2
- Write 0xDEADBEEF at the beginning of SRAM2
- The System Configuration Controller allows to block writing in SRAM2 with a granularity of 1KB. The size of SRAM2 is 32KB. Find the right register to block the SRAM in writing in the first page.
- Try to write a new value at the beginning of SRAM2. What happens?
- Implement the system reset in the HardFaulthandler handler. Read the SCB section in the programming manual to find the sequence to launch an application reset.
- Write 0xAABBCCDD at the beginning of the second page without writing a new value at the beginning of SRAM. Check that there is no application reset.
- Figure out how SRAM write protection can be disabled (see syscfg section in the reference manual?
  Note: Write this entire sequence before the infinite loop and step through the program in debug mode to verify that the requested features work properly.

FLASH WRITE PROTECTION

It is possible to write-protect Flash using option bytes. The way to protect Flash depends on the product (with sector numbers or directly with addresses).

Work to be provided

- ✓ Find the option bytes that allow to protect the FLASH on the STM32L476.
- ✓ Check that it is possible to write a new value on the first word at address 0x0800 0000
- ✓ Using the cubeProgrammer, protect the first section.
- ✓ Try to write a new value on the first word at address 0x0800 0000 and check the error message.
- ✓ Reconfigure the microcontroller again to be able to write to the Flash and confirm that this is indeed the case.